

LM07 Introduction to Digital Assets

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1. Introduction

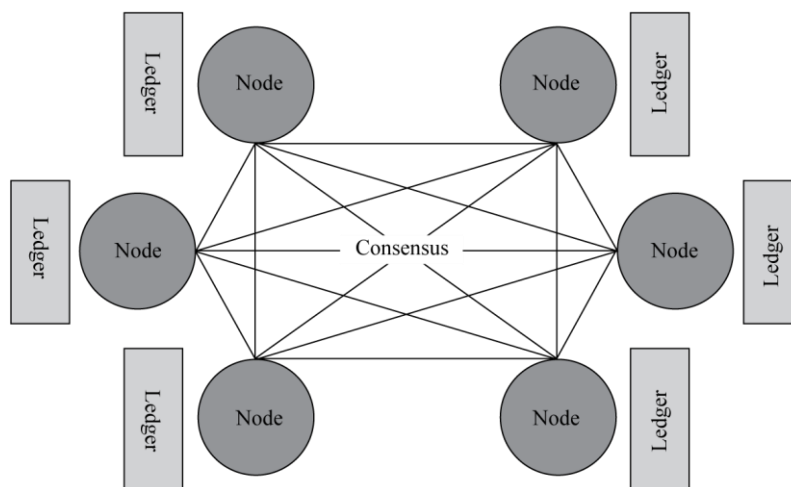
Digital assets are a relatively new investment class consisting of assets that can be created, stored, and transmitted electronically and have ownership or use rights associated with them.

This learning module covers:

- Distributed ledger technology
- Investment features of digital assets
- Investment forms and vehicles used in digital asset investments
- Investment risk, return, and diversification benefits of digital asset investments

2. Distributed Ledger Technology

A **distributed ledger** is a database which can be shared across computer entities (or nodes) in a network. This is illustrated in the following diagram.



There can be a potentially infinite number of nodes in a network. Every node will have a copy of the distributed ledger. There is a consensus mechanism which ensures that all these ledgers are kept in sync. Through the consensus mechanism all nodes agree on a new transaction and update their ledgers. New records are considered immutable, which means once a record is created it cannot be changed.

Distributed ledger technology (DLT) uses **cryptography**, which refers to encrypting and decrypting data. Through encryption, we ensure that the data remains secure.

DLT also accommodates **smart contracts**. These are computer programs that self-execute on the basis of pre-specified terms and conditions. For example, contracts that automatically transfer collateral from the borrower to the lender in the event of default.

DLT networks allow us to create, exchange, and track ownership of financial assets on a

peer-to-peer basis. There is no central authority to validate the transactions.

DLT benefits include:

- Accuracy, transparency, and security in the record keeping process.
- Faster transfer of ownership.
- Peer-to-peer interactions.

A drawback of DLT is that the computational processes underlying DLT require massive amounts of energy to verify transaction activity.

Blockchain is a type of distributed ledger. Its characteristics are:

- Information is recorded sequentially within blocks.
- Blocks are chained and secured using cryptography.
- Each block contains a grouping of transactions and a secure link to the previous block.

The following steps outline the process of adding new transactions to the Blockchain network.

1. A transaction takes place between buyer and seller.
2. The transaction is broadcast to the network of computers (nodes).
3. The nodes validate the transaction details and parties to the transaction.
4. Once verified, the transaction is combined with other transactions to form a new block (of predetermined size) of data for the ledger.
5. This block of data is then added or linked (using a cryptographic process) to the previous block(s) containing data.
6. The transaction is considered complete and the ledger has been updated.

Proof of Work vs. Proof of Stake

A consensus protocol is a set of rules that govern how blocks in a blockchain network are cryptographically chained together for the verification of the complete and immutable history of transaction records.

Consensus protocols are classified into two types: "proof of work" (PoW) and "proof of stake" (PoS).

The Proof of Work Protocol

The proof of work protocol uses a computationally expensive lottery to determine which specific block to add. To verify a transaction, the PoW consensus mechanism employs a cryptographic problem that must be solved by some computers on the network (known as miners) each time a transaction occurs.

Miners use powerful computers and large amounts of energy to solve complex algorithm puzzles in order to validate and lock blocks of transactions into the blockchain and earn cryptocurrency in the process. The "proof of work" consensus process for updating the blockchain can require significant amounts of computing power, making it extremely

difficult and expensive for a single third party to manipulate historical data.

The Proof of Stake Protocol

This protocol requires selected network participants, known as validators, to pledge capital in order to vouch for the validity of a block. This stake signals the network that a validator is available to verify the veracity of a transaction and propose a block.

A majority of the other validators who have staked a digital asset in the network must then attest to the proposed block's validity. Validators benefit from both proposing and attesting to the validity of blocks proposed by other participants.

The security standards of the PoS protocol are based on a group of stakeholders (and their pledged stake) controlling the network's computational power and protecting access from malicious parties gaining a majority.

Permissioned and Permissionless Networks

DLT can take the form of permissionless or permissioned networks.

Permissionless networks are open to any user who wishes to make a transaction. Once a transaction is added, it cannot be changed. All users can see all transactions on the block chain. These networks do not depend on a central authority. Bitcoin is a popular example of an open permissionless network.

In **permissioned** networks, network members may be restricted from participating in certain network activities. Controls or permissions might be used. Different users may have different levels of access to the ledger. For example, participants may be allowed to enter transactions while regulators may be allowed to view the transaction history.

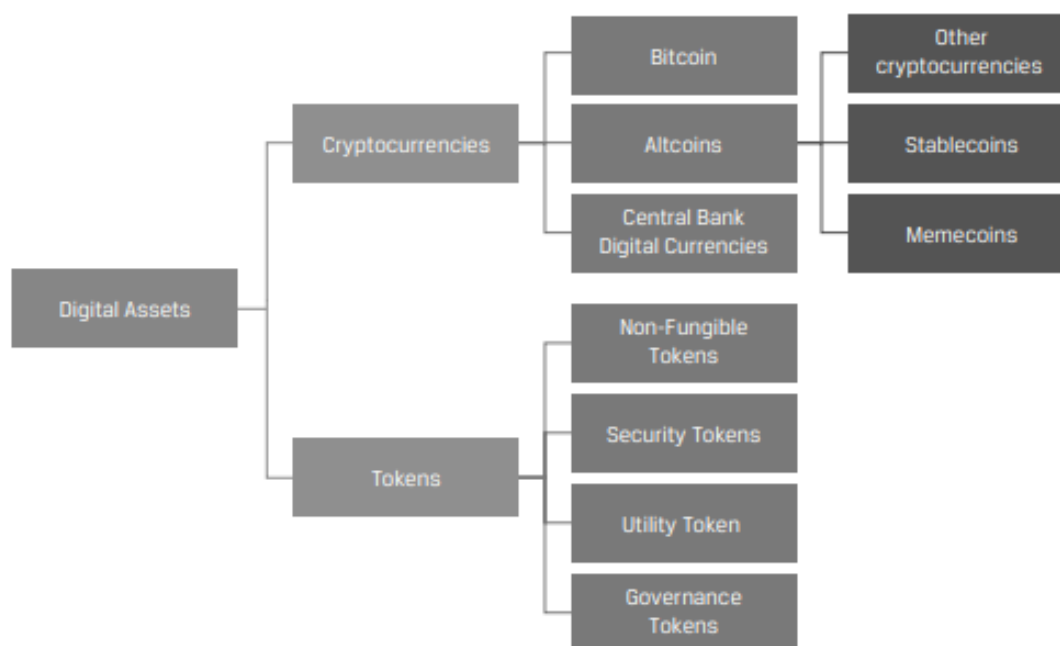
Exhibit 3 from the curriculum compares the features of permissioned and permissionless blockchains.

	Permissioned blockchain	Permissionless blockchain
Speed	Faster as only a limited number of members participate or are authorized to validate transactions	Slower as a large number of members have to reach consensus, which decreases network speed and scalability
Cost	Cost-effective as few members are required to validate each transaction	Not cost-effective as many members are required to validate each transaction
Decentralization	Partially decentralized as there are a limited number of members in the chain	Decentralized as all members can access the network
Access	Membership is limited.	Membership is unlimited.
Governance	The governance is determined by a centralized organization.	The governance is decentralized and is maintained by the members.

Types of Digital Assets

Digital assets are assets that exist only as an electronic record with the right to use, buy, or

sell them. Exhibit 4 from the curriculum shows the different types of digital assets.



Cryptocurrencies:

They are also called digital currency or electronic currency. They do not have any physical form, but allow transactions to take place between buyers and sellers. They are issued by private individuals or organizations. Usually there is no central authority, like a central bank backing these currencies, unless they are ‘central bank digital currencies’.

Many cryptocurrencies have a self-imposed limit on the total amount of currency they may issue. For example, a well-known cryptocurrency, Bitcoin has a self-imposed limit of 21 million. Such limits could help maintain a value from a technical perspective, yet we should also recognize the fact that with cryptocurrencies there is a lack of fundamentals underlying the value of the currency. Hence, they tend to be very volatile relative to major currencies like the Dollar or the Euro.

An *initial coin offering* (ICO) is an unregulated process whereby companies sell their crypto-tokens to investors. Through this process, investors fund the company and the tokens can be used to buy products and services from the company at a latter point in time.

Central banks across the world are recognizing the potential benefits and examining use cases for their own cryptocurrency versions as a substitute to physical currency. Central bank digital currencies (CBDCs) are typically designed as a tokenized version of the central bank's currency ("fiat currency")—essentially, a digital bank note or coin.

Tokenization:

It is the process of representing ownership rights to physical assets on a blockchain or distributed ledger. Usually transactions involving physical assets, such as real estate, require

substantial efforts in ownership verification and examination. DLT can streamline this process by creating a single digital record of ownership.

Another type of digital asset is a **non-fungible token (NFT)**. Using blockchain technology, an NFT connects digital assets to certificates of authenticity. NFTs differ from "fungible" tokens like cryptocurrencies in that each token and the authenticated object it represents are unique. As a result, they can "stamp" assets and represent digital assets in a virtual world. The most common application for NFTs is the trading of digital artwork.

Security tokens digitize the ownership rights associated with publicly traded securities. The custody of these security tokens can be stored on a blockchain, which can increase the efficiency of post trade processing and settlement. The current post-trade clearing and settlement in the financial securities market is quite cumbersome. Security tokens have the ability to streamline this process by providing near-real-time trade verification, reconciliation, and settlement. This can significantly reduce the complexity, time, and cost involved with processing transactions.

Utility tokens provide services within a network, such as pay for services and network fees. While security tokens may pay dividends, utility tokens only compensate for activities on the network.

Governance tokens are important in permissionless networks. They act as a vote to determine how specific networks are run.

3. Digital Asset Investment Features

Digital assets are frequently regarded as an alternative asset class. As digital assets mature, institutional investors may continue to seek exposure to these assets for higher returns and potential diversification benefits.

Distinguishing Characteristics of Digital Assets

Digital assets differ from traditional financial assets in terms of their inherent value, transaction validation approach, uses as a legal medium of exchange, and legal and regulatory protection. Exhibit 5 from the curriculum summarizes these differences.

	Digital Assets	Traditional Financial Assets
Inherent value	▪ No fundamental value or future cash flow generation ▪ Price driven by certain features on the blockchain	▪ Value determined by future cash flow generated from the assets
Transaction validation	▪ Usually recorded on decentralized digital ledgers using cryptography and algorithms for permissionless blockchain networks	▪ Recorded in private ledgers maintained by central intermediaries
Uses as a medium of exchange	▪ Very few digital assets are used as a direct medium of exchange, mainly targeting large-scale commercially viable acceptance	▪ Not used directly as a medium of exchange but can be readily transacted and exchanged into traditional fiat currencies that are widely used in the real world
Legal and regulatory protection	▪ Ambiguous, often contradictory, evolving framework; generally unregulated, with minimal legal protections ▪ Use can be illegal or criminal in some countries	▪ Well-established, tested, and proven legal, regulatory, and commercial standards that are clear, predictable, and well defined across all jurisdictions

Investible Digital Assets

The best-known and widely traded cryptocurrency is 'Bitcoin'. It was designed as an alternative to traditional currencies: as a medium of exchange and store of value.

Several other cryptocurrencies exist on technology similar to Bitcoin. These are called altcoins.

Altcoins

The most popular altcoin is 'Ether'. It was launched in 2015 on its own Ethereum network. It includes an additional feature of a programmable blockchain, which enables users to build applications that use the blockchain to validate or secure transactions. As a result, ether and other programmable coins have the potential to be used for more than just storing value. Such programmable altcoins are also known as smart coins.

Stablecoins

Stablecoins are designed to maintain a stable value by linking their value to another asset and are collateralized by a basket of assets such as fiat currencies, precious metals, or other cryptocurrency. The reserve basket protects holders from price volatility and reduces the risk to stablecoin holders if the cryptocurrency experiences transaction problems.

Meme Coins

Meme coins are cryptocurrencies that are often inspired by a joke and are generally launched for entertainment purposes because they gain popularity quickly, allowing early purchasers to sell their holdings at a significant profit. A popular meme coin is Dogecoin.

4. Digital Asset Investment Forms

Investment in digital assets can be in the form of direct investment on the blockchain or

indirect investments through exchange-traded products and hedge funds.

Direct ownership of Bitcoin and other cryptocurrencies is accomplished through the use of a **cryptocurrency wallet**, which stores the (public and private) digital codes required to access the asset via a computer website or mobile device application.

Cryptocurrency exchanges can be classified into centralized exchanges and decentralized exchanges.

Centralized exchanges:

- Although incompatible with Bitcoin's decentralized ideology, this is the most popular type of exchange.
- Privately held exchanges provide trading platforms for cryptocurrencies and offer volume, liquidity, and price transparency.
- Trading is electronic and direct, with no intermediary broker or dealer, and takes place on private servers, exposing the centralized exchanges and their clients to security risks.
- If the exchange's servers are compromised, the entire system may be rendered inoperable, halting trade, and leaking sensitive user information.
- Some exchanges may be regulated depending on the jurisdiction.

Decentralized exchanges:

- These exchanges mimic blockchain's decentralized protocol and operate similarly to how Bitcoin operates.
- If one of the computers on the network is attacked, the exchange remains operational because there are numerous other computers that continue to operate.
- Therefore, attacking decentralized exchanges is substantially more difficult.
- However, decentralized exchanges are difficult to regulate because no single individual, organization, or group controls the system.

Both centralized and decentralized exchanges are vulnerable to fraud and manipulation, because they are not subject to rigorous oversight and are generally not regulated as financial exchanges.

Direct Digital Asset Investment Forms

Direct investment in cryptocurrencies has several risks:

- The risk of fraud such as scam IPOs, various pump and dump schemes, market manipulation, theft, and schemes that seek to hack access credentials.
- Cryptocurrency wallets can only be accessed with a unique passkey. When the passkey

is lost, the wallet's contents are lost forever. Approximately 20% of all Bitcoins are said to be in lost or abandoned wallets that their owners cannot access.

- Many of the smaller cryptocurrencies may be held primarily by a small group of holders known as "whales." The cryptocurrency community uses the term "whales" to refer to individuals or entities who own a large enough amount of a cryptocurrency to be able to manipulate its price.

Indirect Digital Asset Investment Forms

Alternatives to gain indirect exposure to digital assets include:

Cryptocurrency coin trusts:

- These trusts hold large pools of a cryptocurrency and allow investors to trade in shares of the trust (like a closed-end fund).
- Pros: The investor does not need to create a separate digital wallet. Also, the trust may provide transparency into trading.
- Cons: Trusts charge substantial fees, sometimes more than 2%. The trust shares may trade at a premium or discount to their NAV.

Cryptocurrency futures:

- These are agreements to buy or sell a specific quantity of a cryptocurrency at a specified price on a particular date. Ex: Bitcoin futures on Chicago Mercantile Exchange (CME).
- These contracts are typically cash settled with no actual cryptocurrency changing hands.
- Cons: Futures contracts are inherently leveraged. The cryptocurrency futures market may be less developed, less liquid, and more volatile than more established futures markets.

Cryptocurrency exchange-traded funds:

- Many ETFs that seek to replicate the returns of digital assets have recently been launched.
- These ETFs typically do not invest directly in cryptocurrencies and instead gain exposure to the value of cryptocurrencies through the use of cash and cryptocurrency derivatives.

Cryptocurrency stocks: Examples include equity in publicly traded –

- digital exchanges.
- payment providers accepting cryptocurrencies.

- corporations accepting cryptocurrencies as payments, investing in cryptocurrencies, or mining cryptocurrencies.
- corporations developing and/or manufacturing products or services that are used for running blockchain networks, such as specialized computers used for mining.

Hedge funds investing in cryptocurrencies:

- Several discretionary long, long/short, quantitative, and multi-strategy funds provide exposure to cryptocurrencies.
- Some hedge funds actively mine for Bitcoin to generate further returns.

Digital Forms of Investment for Non-Digital Assets

Asset-backed tokens are digital claims on physical assets, financial assets, or financial instruments and are collateralized by these underlying assets. Examples of tokenized assets include: gold, crude oil, real estate, and equities.

Asset-backed tokens have the potential to increase liquidity by allowing for fractional ownership of high-priced assets such as houses, art, precious metals, and precious stones, allowing multiple investors to own a fractional interest in the same asset. The digital representation of ownership allows for an immutable record of ownership information and ownership transfer, increasing transparency and lowering transaction, intermediation, and record-keeping costs.

Asset-backed tokens are often issued on the Ethereum network or other smart contract platforms that enable peer-to-peer interaction via interoperable, transparent smart contracts that last for the duration of the chain. These decentralized applications, or dApps, enable transactions to occur and be recorded on the blockchain without the need for a central coordinating mechanism.

The push for financial decentralized applications based on opensource codes and smart contracts has grown into a movement known as decentralized finance, or DeFi. DeFi seeks to design, combine, and develop decentralized financial applications as building blocks for sophisticated financial products and services.

5. Digital Asset Investment Risk, Return, and Diversification

Digital Asset Investment Risks and Returns

Bitcoin and other digital assets are priced based on future asset appreciation rather than any underlying cash flow. The market demand for the limited supply of cryptocurrencies is a significant driver of prices. For example, the supply of Bitcoin is limited to 21 million Bitcoins, by design. Due to this some investors view Bitcoins as the digital version of gold.

Bitcoin's performance has been characterized by high return, high volatility and low correlations with traditional asset classes. Bitcoin's price rose from \$0.05 at its inception to a historical high of \$68,789 on 10 November 2021, before crashing to around \$17,709 on 18

June 2022. Early Bitcoin investors were rewarded handsomely with phenomenal price appreciation.

However, as the brief track record shows cryptocurrencies can be very volatile and risky investments. Several countries have imposed severe restrictions on trading and ownership of cryptocurrencies, ex: China, which banned the asset in 2021.

Diversification Benefits of Digital Asset Investments

Because of their historically low correlations with other asset classes, digital assets can help diversify a traditional asset portfolio. However, correlations have been observed to increase, particularly during periods of high market uncertainty.

Summary

LO: Describe financial applications of distributed ledger technology.

A distributed ledger is a database which can be shared across computer entities (or nodes) in a network.

DLT networks allow us to create, exchange, and track ownership of digital assets on a peer-to-peer basis. There is no central authority to validate the transactions.

DLT could also bring efficiencies to post-trade and compliance processes through automation, smart contracts, and identity verification.

DLT can take the form of permissionless or permissioned networks.

A consensus protocol is a set of rules that govern how blocks in a blockchain network are cryptographically chained together for the verification of the complete and immutable history of transaction records. Consensus protocols are classified into two types: "proof of work" (PoW) and "proof of stake" (PoS).

LO: Explain investment features of digital assets and contrast them with other asset classes.

The different types of digital assets are shown in the figure below:

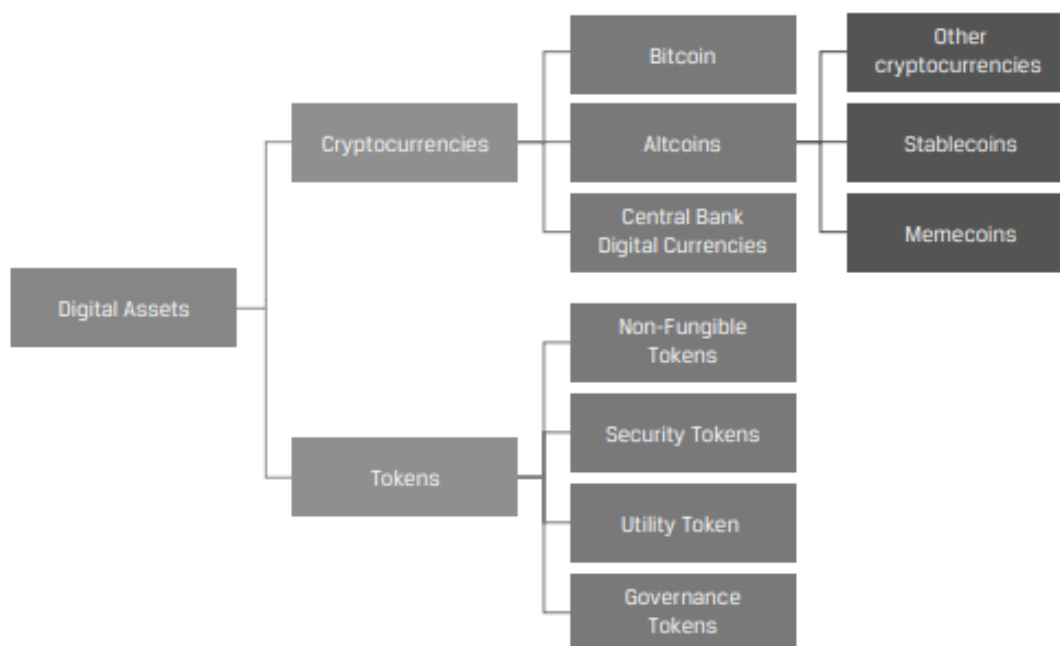


Exhibit 5 from the curriculum summarizes the differences between digital assets and traditional financial assets.

	Digital Assets	Traditional Financial Assets
Inherent value	▪ No fundamental value or future cash flow generation ▪ Price driven by certain features on the blockchain	▪ Value determined by future cash flow generated from the assets
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LO: Describe investment forms and vehicles used in digital asset investments.

Investment in digital assets can be in the form of direct investment on the blockchain or indirect investments through exchange-traded products and hedge funds.

Direct ownership of Bitcoin and other cryptocurrencies is accomplished through the use of a cryptocurrency wallet. Cryptocurrency exchanges can be classified into centralized exchanges and decentralized exchanges.

Alternatives to gain indirect exposure to digital assets include:

- Cryptocurrency coin trusts
- Cryptocurrency futures
- Cryptocurrency exchange traded funds
- Cryptocurrency stocks
- Hedge funds investing in cryptocurrency

Asset-backed tokens are digital claims on physical assets, financial assets, or financial instruments and are collateralized by these underlying assets.

The push for financial decentralized applications based on opensource codes and smart contracts has grown into a movement known as decentralized finance, or DeFi. DeFi seeks to design, combine, and develop decentralized financial applications as building blocks for sophisticated financial products and services

LO: Analyze sources of risk, return, and diversification among digital asset investments.

Bitcoin and other digital assets are priced based on future asset appreciation rather than any

underlying cash flow. The market demand for the limited supply of cryptocurrencies is a significant driver of prices.

Bitcoin's performance has been characterized by high return, high volatility and low correlations with traditional asset classes.

Because of their historically low correlations with other asset classes, digital assets can help diversify a traditional asset portfolio. However, correlations have been observed to increase, particularly during periods of high market uncertainty.